

RNPM: Neural-Guided Embedding Region Selection and Error Correction for Robust Audio Multi-Watermarking

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Abstract

Audio multi-watermarking demands inserting multiple independent payload keys (e.g., copyright ID, timestamp, user signature) simultaneously without degrading audio fidelity. This study introduces RNPM, a neural-guided region selection network that identifies perceptually insensitive regions in audio Short-Time Fourier Transform (STFT) spectrograms, combined with BCH error-correcting codes for resilient extraction.

Key Methodology & Architecture

STFT spectrogram analysis, Neural Saliency Masking network, BCH Error-Correcting Code, multi-key orthogonal embedding.

Datasets & Benchmark Environments

LJSpeech-1.1, LibriTTS, ESC-50 Environmental Audio.

Performance Metrics & Graph Axis Parameters

Reported Results: Payload Capacity = 128 bits/sec, PSNR = 44.8 dB, BER under 15 dB Gaussian noise = 0.02%.

Graph Parameter Mapping: X-axis: Watermark Capacity (bits per second, 16 to 256 bps); Y-axis: Perceptual Audio Quality (ODG Score, -4 to 0).

Comparative Analysis

Advantages (Pros)	Limitations (Cons)
Supports multiple concurrent watermark payloads with low bit error rate.	STFT framing artifact potential if frame size is misconfigured.

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